

Multiscale haptic interfaces for metaverse

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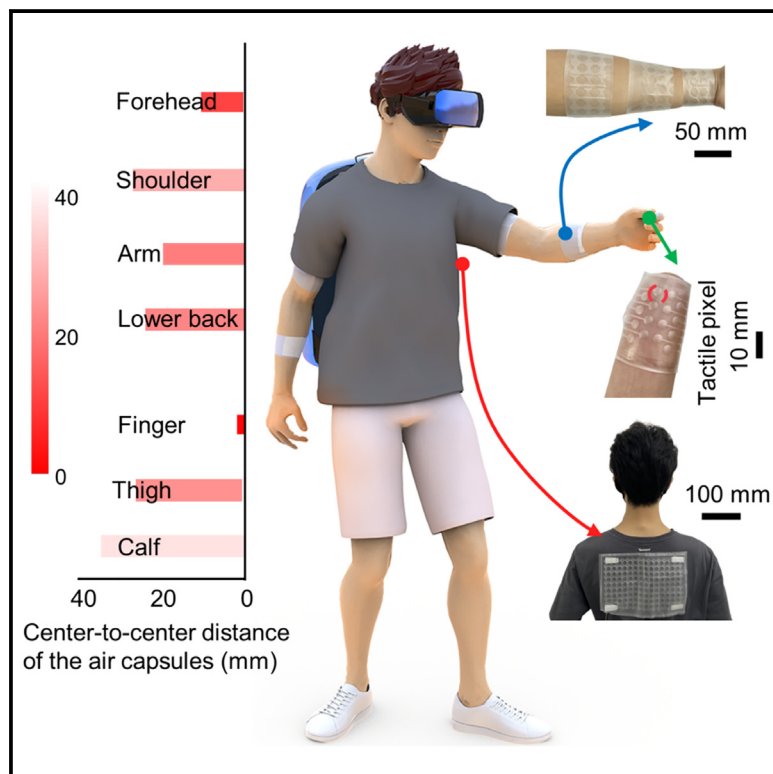
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Multiscale haptic interfaces for metaverse

Graphical abstract



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In brief

A flexible, thin, and lightweight pneumatic haptic interface with customizable dimensions and resolution is presented. The pixel-like and patch-like air capsules are fabricated using a one-step hot-pressing process, which can be scaled up for mass production. Application demonstrations include braille reading and other interactions in virtual reality.

Highlights

- A flexible pneumatic haptic interface is developed for virtual reality applications
- The air capsule layout can be configured for the local sensitivities of the body
- The cheap and scalable manufacturing process is suitable for mass production
- Braille recognition and interactions with a virtual pet are demonstrated



Develop

Prototype with demonstrated applications in relevant environment

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Article

Multiscale haptic interfaces for metaverse

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THE BIGGER PICTURE Haptic systems can expand the immersive experience in virtual reality. However, compared to video and audio interfaces, haptic interface devices are relatively underdeveloped with the limitations of low fidelity, large sizes, and high manufacturing costs. In this study, we present a multiscale pneumatic haptic interface (PHI) with inflatable thermoplastic urethane capsules that are soft, thin, and easy and cheap to manufacture by hot pressing. The two different designs of the PHIs—one for a pixel-like surface suitable for the fingertips and one for a patch-like surface for the arm and the body—can be used as a blueprint for designing multiscale haptic interfaces for multimodal virtual reality experiences in the future.

SUMMARY

Haptic interactions in virtual reality (or the “metaverse”) depend on the ability of actuators to transmit sensations to the mechanoreceptors on the human skin. Similar to the considerations for foveated display, variations in the sensitivity of our somatosensory system are important to consider when designing haptic devices, i.e., to design a system with adjustable actuator size and density to adapt for the different sensitivities and distributions of mechanoreceptors on the human body. In this paper, we present a wearable haptic feedback device made of thermoplastic urethanes with an adjustable pneumatic haptic interface (PHI). The PHI contains actuator types of different sizes and layouts to provide haptic feedback ranging from delicate and precise feedback to robust and large-area feedback. We develop a 16-channel system with a refresh rate up to 10 Hz and demonstrate its applications in gaming and training programs in the metaverse and for braille reading.

INTRODUCTION

The human skin can perceive a wide range of sensations such as mechanical^{1,2} and thermal^{3,4} stimulations of different spatial scales and timescales, e.g., touching a large, cold glass door versus being poked by a small, hot metal fork. These sensations are of paramount importance for creating an immersive experience in virtual reality (VR; also known as the “metaverse”). Reproducing this tactile feedback is challenging, as it requires high-fidelity, bio-friendly, lightweight, and large-sized tactile devices.^{5–8} However, efforts to integrate mechanical programmable interfaces for a metaverse platform are in their relative infancy compared to those of video and audio interfaces.^{9,10} There are mainly two design strategies: mechanotactile actuators that use linear motors,¹¹ electromagnet actuators,^{12–14} dielectric

elastomers,¹⁵ and shape memory alloy¹⁶ to generate mechanical feedback and electrotactile actuators^{17–19} that use electricity to emulate the same sensations. Pneumatic haptic actuators, which are a type of mechanotactile actuator, have the advantage of having similar elastic moduli to skin,^{20–27} but their fabrication using elastic materials such as polydimethylsiloxane, Ecoflex, and Sylgard, tends to be more complex.

While many research works have focused on improving the density and performance of tactile actuators,^{17–19,21,23–25,27} less attention has been paid to creating a large-scale haptic interface with adaptable actuator density. The ability of a haptic interface to cover a larger surface area of skin implies an increased capacity to furnish haptic information regarding spatial location. However, the distribution of tactile receptors on human skin varies greatly, with a much higher density of



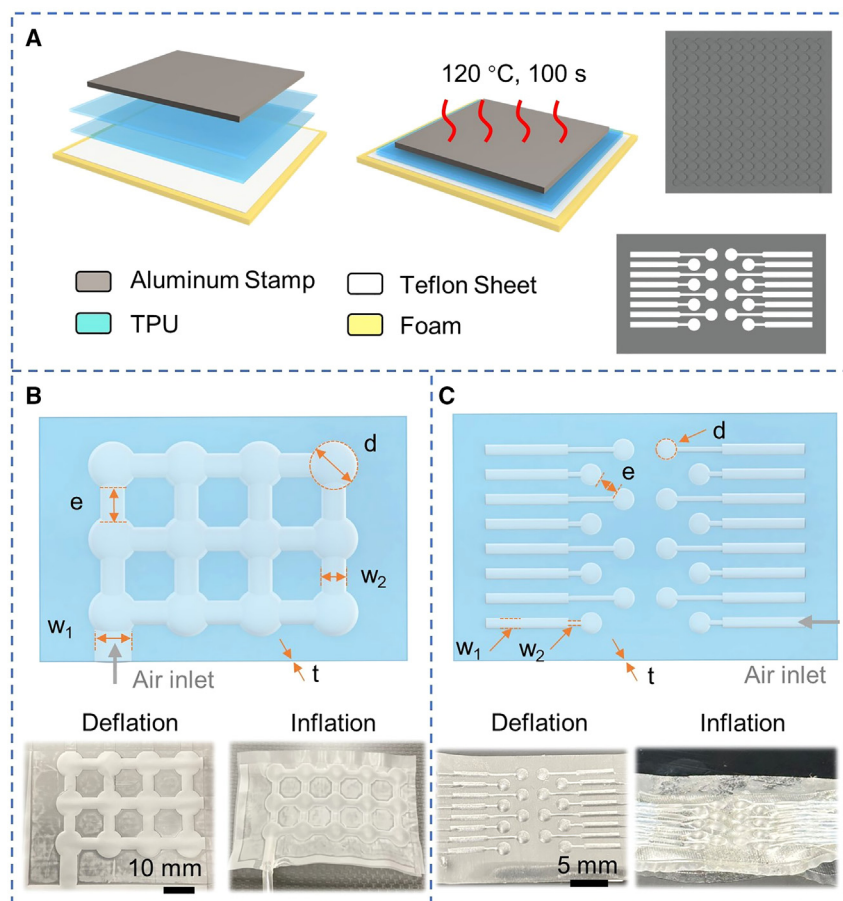


Figure 1. Fabrication and geometric dimension of the PHI

(A) Illustration of one-step manufacture. The PHI is fabricated from TPU films sealed by heat using a machined metal stamp. The processing temperature and time are set to 120°C and 100 s, respectively. The left stamp with a groove is for the patch-like PHI, while the stamp with holes is for the pixel-like PHI.

(B and C) Design and optical images of patch-like PHI (B) and pixel-like PHI (C) with dimension labels. The diameter of the balloon is designed according to the two-point discrimination and PHI characterization metal stamp of PHI.

120°C and then pressing the metal stamp, two pieces of TPU film, Teflon sheet, and foam together. The foam helps distribute the pressure between the TPU films and metal stamp more uniformly, while the Teflon sheet prevents the TPU film and foam from sticking together. After 100 s of hot pressing, the two TPU films are tightly bonded. The grooves and holes in the stamp do not apply pressure on TPU films, creating the air channels and capsule chambers.

Figures 1B and 1C show the two different layouts for the PHIs, either of which can be made in different sizes for use in different parts on the body. The patch-like PHIs (Figure 1B) have a lattice of connected air capsules for providing haptic feedback for a required area, while

tactile receptors on the fingertips (241 unit/cm²) than on the palm (58 unit/cm²) or thigh (9 unit/cm²).²⁸ Here, we present a pneumatic haptic interface (PHI) system made of inflatable air capsules of different sizes and layouts that can be adapted for providing the desirable haptic feedback at different spatial resolutions. The PHI is made of elastic and tough thermoplastic urethanes (TPUs),²⁹ is easy to manufacture, and can be inflated using a pump and self-deflate rapidly.

RESULTS

System design and characterization

PHI design and fabrication

The PHI is made of two layers of TPU films, which offer advantages such as high elasticity, non-toxicity, high fatigue resistance, and soft texture.^{30,31} Compared to silicone rubbers, TPU does not require processing steps like stirring, vacuuming, or heat curing. The shape of TPU can be adjusted by simply applying heat and pressure and then letting it cool to retain its shape. TPU can also undergo large deformations with a high shape-recovery feature.³²

The fabrication process of the PHI involves only a one-step process, i.e., hot pressing (see Figure 1A). Custom aluminum alloy plates are used to stamp out the designed geometric size. The fabrication process involves setting the hot press machine to

the pixel-like PHIs (Figure 1C) contain separate air capsules that can be independently controlled to provide a higher resolution. The pixel-like PHIs can be arranged symmetrically along the contour of the finger to provide haptic spatial localization.³³ The geometric dimensions designed based on the experiments in the PHI characterization section are listed in Table 1. Both PHI designs are inflated through an inlet neck where a polyurethane (PU) tube can be inserted using an adhesive. After inflation, the reserved air capsule chambers expand into a hemispherical shape (Figures 1B and 1C).

For both designs, the width of the air inlet channel (w_1) was chosen to be 1.2 times the outer diameter of the PU tube to ensure that the tube can be inserted into the PHI. The width of the airway (w_2) is a critical parameter that affects the deformation and the resistance of the airway.³⁴ For PHIs with a denser air capsule distribution, the airway needs to have a smaller width while meeting the millisecond response time of air capsules. According to the finite-element analysis (FEA) results (Figure S1), the parameters of the pixel-like PHI and patch-like PHI were designed to be 0.8 and 4 mm, respectively. The array spacing parameters (d , e) are chosen to align the distribution of air capsules with the requirements of the two-point discrimination test. Once the diameter (d) is determined, the suitable edge-to-edge distance (e) can be selected within the corresponding range based on the manufacturing process.

Table 1. Geometric dimensions of PHI

Symbol	Description	PHI on fingertip	PHI on body
d	air capsule diameter	2.5 mm	8 mm
t	the thickness of each layer of PHI	70 μm	140 μm
w_1	width of the air inlet channel	2.2 mm	4.5 mm
w_2	width of the air channel	0.8 mm	4 mm
e	edge-to-edge element spacing parameter	1.5 mm	>2 mm ^a

^aDepends on the required stimulation density at the placement location.

Previous tactile arrays for fingertips range center to center (i.e., $d + e$ in terms of the terminology in this paper) between 0.1 and 3.5 mm.^{11,13,17,27,35} In comparison, the general size of a braille dot is about 1–2 mm with a center-to-center distance of 2–4 mm between adjacent dots.³⁶ During the PHI fabrication, three different values of edge-to-edge distance e were tested: 1.0, 1.25, and 1.5 mm. Spacing parameters of 1.0 and 1.25 mm spacing exhibited occasional edge disassociation from the TPU film due to the limited contact area between channels. The spacing parameter of 1.5 mm does not suffer from the above problems, making it a more suitable value to balance size and reliability.

PHI characterization

To optimize the PHIs for use in the metaverse, static and dynamic performances were studied using a series of air capsule diameter tests, dynamic response tests, and endurance tests. Air capsule diameter tests were optimized to match the sensitivity of human fingers and body while still being compact enough to form a functional array. Dynamic response tests were devised to examine the fastest refresh rate of the control unit that can supply enough force on the air capsules. Endurance tests were performed to study the reliability of air capsules with different diameters and thicknesses (t). The air capsules are actuated by inflating them with air from a pressurized air source and deflating by passively venting the air.

A laser displacement sensor (CL-3000; Keyence) and a high-precision 3D force sensor (Nano 17; ATI Industrial Automation) were used to measure the deflection and force of PHI, respectively (Figures 2A and 2B). The air pressure inside the PHI was adjusted using a pressure regulator and an on-off valve. Individual air capsules with different diameters were tested with five air pressure configurations. Each configuration was tested 10 times. Figures 2A, 2B, and S2 show the experiment results, which reveal that the measured deflection and force output are larger with larger air capsules and thinner films. To maintain consistent experimental conditions, the same probe with a diameter of 2.6 mm was used to test air capsules of different sizes. The 30- and 70- μm -thick PHIs had similar deflection and block force, while the 100- μm -thick PHI had smaller outputs. Because of this, the 70 μm design was chosen for reliability over the 30 μm design. To maximize the number of air capsules on the fingertip and in combination with the minimum spacing parameters determined above, a

PHI with a diameter of 2.5 mm and a thickness of 70 μm was considered. For the patch-like PHIs, the 120- μm -thick air capsule fails at 80 kPa, while the deflection and block force of the 180- μm -thick air capsule is too low. The deflection and force of 140- and 160- μm -thick air capsules were tested. As shown in Figures 2A, 2B, and S2, the outputs for the two parameters ($t = 140 \mu\text{m}$, $d = 8 \text{ mm}$ and $t = 160 \mu\text{m}$, $d = 10 \text{ mm}$) show similar performance. The final size selection is further optimized by considering the endurance tests.

Endurance tests were conducted on the differently sized air capsules. Each parameter combination was tested at a pressure of 100 kPa for more than 3,600 cycles, with each consisting of 0.5 s of inflating and 0.5 s of deflating. The deflection of 70 μm /2.5 mm and 140 μm /8 mm (t/d) air capsules remained consistent over the 3,600 actuation cycles (Figure 2C). However, as shown in Figure S3, the deflection of the 160 μm /10 mm air capsule increased with the number of execution cycles. This can be attributed to the deformation of the TPU films when stretched and the layers separating under pressure. In conclusion, the thickness and optimal diameter combinations of air capsules are set as 70 μm /2.5 mm and 140 μm /8 mm for the pixel-like PHI and patch-like PHI, respectively.

Threshold tests were conducted to determine the minimum pressure supply at which individuals can feel the deformation of the PHI air capsules. 20 volunteers were invited to participate in the experiment, during which they wore PHI on their fingertips. The output pressure of the air pump was regulated, and the PHI air capsules were randomly inflated. The pressure at which the volunteers could feel the deformation was recorded (Figure S4). With the chosen 70 μm /2.5 mm configuration, four volunteers were able to feel the deformation at 20 kPa, while six volunteers could detect the deformation at 10 kPa. As a result, we consider 20 kPa as the minimum working pressure for the PHI system.

To validate the experimental measurements, we established an analytical model that included material property measurements and FEA. We used a tensile tester system (TY8000-A, Tian Yuan Test Instrument) to measure the Young's modulus of TPU. The geometric size and optical image of the tensile sample are shown in Figure S5A, and the size of the test area is 40 $\times$ 5 $\times$ 0.3 mm. Figure S5B illustrates the stress-strain curve of the tested tensile samples. By calculating the stress-strain ratio in the linear region, the Young's modulus of TPU is about 18 MPa. To theoretically investigate the relationship between inflation pressure and maximum height during inflation, an FEA simulation based on COMSOL Multiphysics is carried out. As shown in Figure S5C, the air capsule (70 μm /2.5 mm) becomes hemispherical after pressure is applied. A boundary point probe was set on the center of the air capsule to record the maximum height during the increasing inflation. The experimental values of PHI under different given inflation pressures closely matched the expected geometrical model (Figure S5D).

PCWPS system design

Figure 3A shows the diagram of a fully programmable closed-loop wearable pneumatic haptic system (PCWPS) consisting of a microcontroller unit, a battery, metal-oxide-semiconductor field-effect transistor (MOSFET) circuits, air pressure sensors, solenoid valves, and a diaphragm pneumatic pump. The system is divided into three pathways: control signal (green line), airflow (blue line),

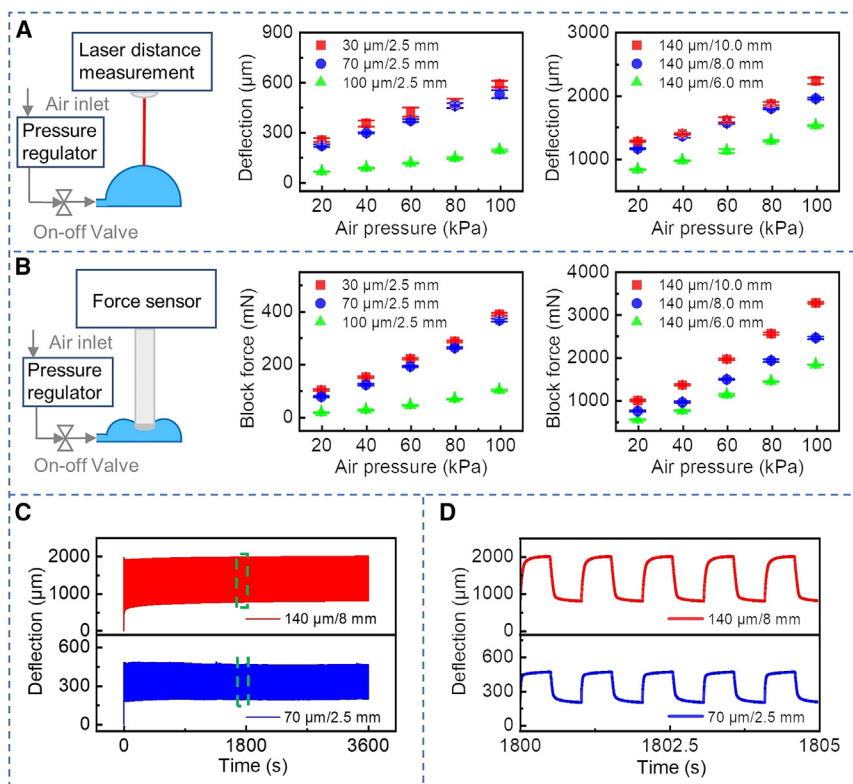


Figure 2. Characterization of the PHI

(A) Experimental setup and deflection response of each geometric configuration under different pressures. Data are shown as mean $\pm$ SD ($n = 10$). (B) Experimental setup and block force response. Data are shown as mean $\pm$ SD ($n = 10$). (C) Endurance test result of PHI over 3,600 cycles. The green dotted box shows the zoomed-in image.

which cannot provide enough air in time to support the expansion of the air capsules.

Figure 3C illustrates the force output for three of the test points. At point 1, the force output can maintain the maximum value, while at point 2 and point 3, the output would hit a spike and then lowered with increasing refresh rates and channels. The reason for the spike is that after the solenoid valve is opened, the high-pressure gas stored in the manifold flows quickly to the PHIs along the tube, causing the force response in the air capsule to sharply increase. However, the flow rate generated by the air pump for a short time cannot maintain the pressure in the entire tube, resulting in a drop in the force output.

and power supply (red line). The microcontroller unit is controlled by the host computer through a serial interface universal asynchronous receiver/transmitter and generates digital control signals to the MOSFET circuits, which regulate the on-off frequency of the solenoid valves. The airflow generated by the air pump (100 kPa, 15 L/min) passes through the solenoid valves and is distributed to the 16 channels. Each channel is connected to a PHI unit and a pressure sensor by a PU tube. Air capsules are inflated under the pressure while the sensors transmit the pressure signal of each channel to the microcontroller and pass to the host computer. The pressure sensor signals are digitized and transmitted to a proportional-integral (PI) control algorithm, which is implemented in the host computer. The PI control component is added to the pressure set point after being multiplied by a pre-determined gain (Figure S6). The portable battery, which has a capacity of 82 Wh, can output 24 and 12 V at the same time. The working time was conducted under the condition that the air pump continuously operated all 16 solenoid valves at a frequency of 1 Hz. The working times of different solenoid valve duty cycles are shown in Figure S7. The working time of the system at maximum power is 2.22 h.

There were 25 combinations of different refresh rates and numbers of channels (see Figure 3B) that were measured to plot the force-frequency response map, where the color gradient indicates the force output of PHI detected by the sensor. Each channel was connected to a piece of PHI that has 15 air capsules (140 μm/8 mm) through a 1-m-long PU tube. As the number of channels and the frequency increase, the blocking force tends to decrease. This is due to the performance of the air pump,

The air pressure response map shows a similar tendency to the force-frequency response map (Figure S8). Moreover, there are fluctuations at the beginning of each period in the graph of a particular point, which agrees well with the force-frequency response map. The highest refresh rates are 10 Hz when 16 channels of PHI are driven and 10 Hz when 12 channels of PHI are driven, given that the minimum operating pressure of the system is 20 kPa. The fitted line (red dash line) is shown in Figure S8. Additionally, the system response time is about 30 ms.

Application demonstrations for PCWPS

The mechanoreceptors distributed across the skin, within the dermis, are responsible for measuring mechanical stimuli. However, the density of mechanoreceptors varies significantly across the skin, with the fingertips having the highest density and the calf having the lowest.^{37,38} To optimize the use of tactile displays for information communication, perceptual resolution and sensitivity must be taken into consideration. Due to the low refresh rate of the pneumatic tactile feedback system, the static two-point discrimination test results were selected as the basis for the air capsule size design. Smaller air capsule spacing allows for richer tactile feedback but also results in larger driver sizes. The spacing based on two-point discrimination tests enables discernible tactile feedback while reducing the number of air channels and minimizing device complexity. Stevens et al.³⁹ conducted the two-point discrimination and pressure sensitivity tests, which show that there are non-negligible gaps in resolution between different parts of body. Therefore, the size of the actuator and the magnitude of the output

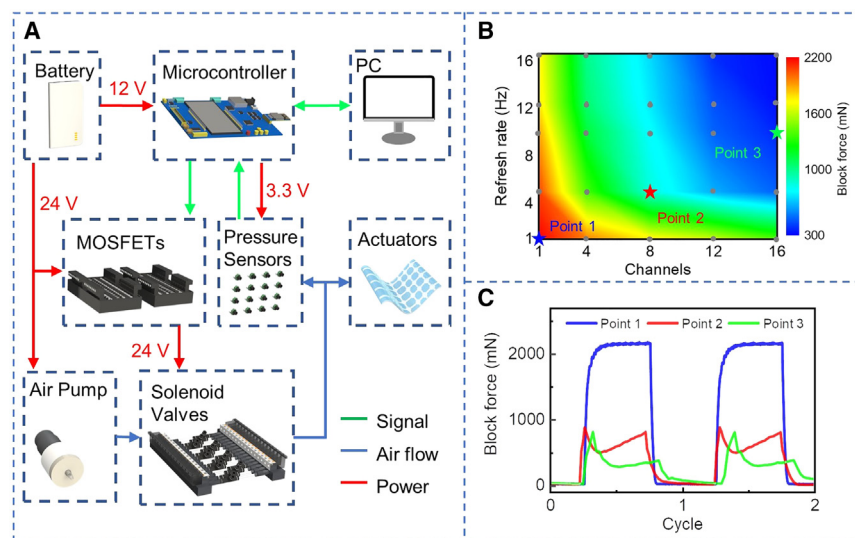


Figure 3. Schematic diagram and dynamic response

(A) Illustration of PCWPS. The whole system is battery powered and receives control signals from the computer. Each channel can be independently controlled and has a pressure sensor to measure the air pressure.

(B) Experimental results (gray dots) for force response with different refresh rate and channel combinations. The tube is 1 m long, and the pressure is 100 kPa. Original data can be found in Table S1.

(C) Force response of three specific points.

force need to match the sensitivity of the deployed area. By adjusting the diameter and the edge-to-edge distance between air capsules, the stimulus distribution can perfectly match the two-point discrimination results.

Due to its elasticity and flexibility, the PHI can be attached to most parts of the body, including the limbs, fingers, and torso, using double-sided tape or Velcro. Each unit of the PHI can be inflated and deflated by the PCWPS, and the number of units depends on the number of solenoid valves in the PCWPS. Figure S9 shows a volunteer wearing the metaverse device, which includes a VR headset, a tracker, a backpack containing the PCWPS (weight 3.5 kg), and PHIs on the body. In addition, PHIs can also be attached to rubber finger cots, sleeves, or clothing to achieve tighter attachment to the skin (Figure S10).

Braille recognition tests

To demonstrate the potential of the PHI in providing a delicate sense of touch, we performed a braille recognition test. Unlike other tactile rendering methods that use vibration and electro-tactile effects to stimulate the fingertip,^{13,40} our PHIs provide a deformation similar to real braille letters. Figure 4A shows the testing environment where a volunteer placed the tip of their index finger on the PHI to decipher the braille letter rendered by the PHI (Video S1). Braille is a pattern of six raised or non-raised dots. To recognize braille letters, people typically run their fingers over the dots, feeling for the raised dots that make up the individual letters. By using air capsules in different areas of the PHI, the sensation of sweeping a finger on a braille letter can be emulated by the sequential inflation of the individual PHI pixels (Figure 4B).

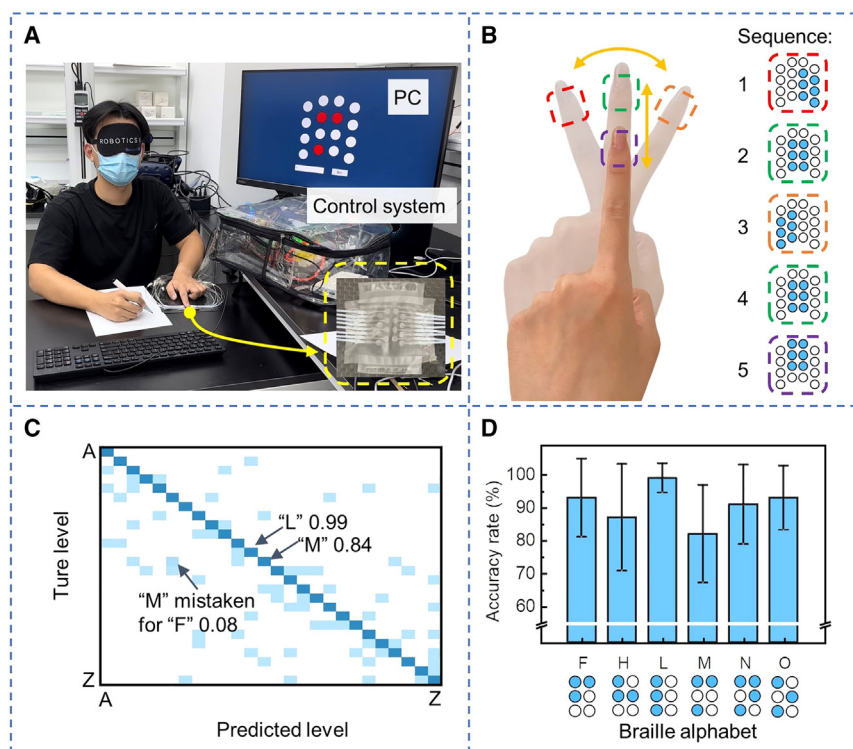
We tested the braille alphabet letter by letter. For each letter, 20 volunteers participated in 5 sets of tests resulting in 100 sets of data. During the test, the volunteers did not know the real pattern in advance. They were tasked to feel the pattern with their fingertips and then write it down on paper. The test results showed a recognition accuracy of over 90% (Figure 4C). Analysis of the confusion matrix revealed that misidentification mainly occurred among letters within the “MF”, “QN,” and “ZO” sets (the detailed confusion matrix is shown in Figure S11).

As expected, the simpler the pattern was, the higher the accuracy of recognition. For pattern recognition of the letter “M,” the accuracy was only 84%. For the letter “L,” which only uses a row of three air capsules, the accuracy was 99%. Numerous identification errors were due to user misinterpretation in the vertical direction. For example, pattern “A”-“I” only uses the top four dots, which were often identified as the lower four dots in the test process (Figure S12). Identification accuracy can be improved by optimizing the relative location according to personal preferences. These test results demonstrate that the PHI is sufficient for braille recognition based on sensing by fingertip areas and will provide great convenience for communication among people with visual impairments.

VR interaction demonstrations

The PHI has the capability to translate virtual signals into tactile sensations, making it an ideal technology for enhancing tactile experiences during interactive entertainment in the metaverse. Due to its flexibility, thinness, and electrodeless features, the PHI can be placed on any area of the human body without causing a pulling sensation due to movement, a weight-bearing sensation due to wearing the device, or insecurity due to fear of electric shock. With custom aluminum stamps, the PHI can be scaled up and used in entertainment scenarios that require the combination of multiple body parts.

Figure 5A and Video S2 demonstrate the potential of using a PHI for interacting with a cat in the metaverse. In this scenario, a volunteer wears the pneumatic-tactile rendering system to enter a virtual world and interact with a virtual cat. Three PHI combinations (t/d), 70 $\mu\text{m}/2.5$ mm, 140 $\mu\text{m}/6$ mm, and 140 $\mu\text{m}/8$ mm, were attached on user’s fingertip, palm, and arm, respectively. The edge-to-edge distances (e) are designed to be 1.5, 5, and 6 mm to match tactile sensitivity. By inflating and deflating the air capsules on the fingertip in sequence, the PHIs simulate the sensation of being licked by the cat. Combined with the visual feedback of the cat interacting with the volunteer, the PHI attached to the arm can also be inflated and deflated to simulate the pressure created on the arm by the cat. The tactile pressure profiles that PHIs exert on the skin are recorded by a flexible sensor (FLX, Matrix Innovation). The pressure map matches the distribution of the air capsules. This demonstrates the potential for PHIs to be used in a wide range of VR applications,



media entertainment experiences, professional training, and educational feedback.

However, there are some limitations that need to be addressed. The current form factor and weight of the PCWPS are limitations but can be optimized by customizing miniaturized components and replacing metal with plastic materials. The refresh rate of PCWPS is another limitation caused by long inflation and deflation times. This is because of the large resistance in the air tube and the insufficient air pump performance resulting in the inability to inflate and deflate quickly. Optimizing the diameter of the air tube and using a high-performance

air pump can improve the refresh rate of the PCWPS. High-frequency electrical stimulations can make up for the inability of pneumatic systems to provide high-frequency tactile feedback.

Printing a layer of electrodes on the PHI to combine pneumatic mechanical force feedback and high-frequency electrical stimulation may be a potential solution. The PHI presented here contains two layers of TPUs of the same thickness, and the outer layer facing away from the skin is modestly deformed while providing no useful information to the user. We tried using different materials on the back as well as thicker TPU, but the bonding did not work well. In the future, the processing technology can be optimized to improve the reliability of PHIs with different back sides. Different display patterns could also be explored to better integrate more independently controllable air capsules. While adding more solenoid valves can be a solution, it also increases the size and weight of the system. Introducing pneumatic circuits to reduce the number of airways required may also be a solution.⁴¹

DISCUSSION

In conclusion, thin, light, flexible, and multilevel sizes adapted to human tactile sensitivity enable PHIs to produce more immersive haptic feedback on the body. Coupled with its simple fabrication, the PHI has the potential to become a platform technology for applications requiring immersive VR in combination with whole-body-tactile-rendering, comfortable actuators.

Compared to other mechanical tactile actuators, such as electromagnetic and pneumatic stimulators, the PHI is a soft and thin wearable platform that can directly laminate onto the skin to provide continuous force feedback by stimulating the mechanoreceptors through deformation. Two schemes were designed for the fingertip and the body to accommodate different levels of tactile sensitivity. The one-step manufacturing process greatly reduces production time and cost, making PHIs a promising solution for commercialization. Demonstrations of Braille recognition and interaction in the metaverse show the potential for PHIs in applications that require more intuitive and comprehensive tactile interactions, including communication systems, multi-

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests should be directed to and will be fulfilled by the lead contact, Zhengbao Yang (zbyang@ust.hk).

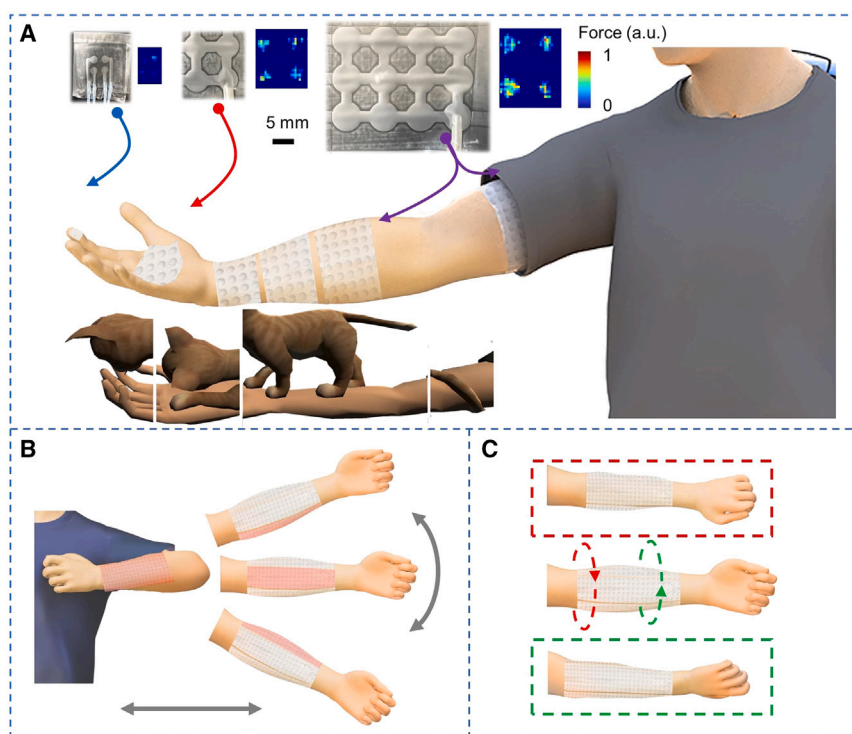


Figure 5. Entertainment and action guide

(A) Layout plan of PHI in interaction game with a cat. PHIs on the fingers, palm, forearm, and upper arm are inflated in sequence to simulate the sensation of a cat's tongue licking, cheek touching, foot stamping, and tail winding. (B) Strategy of swing motion. The red PHI is inflated to prompt the user to swing their arm in a specific motion. (C) Strategy of rotation motion. The user is prompted by the air capsules in which direction they should rotate their arm.

Materials availability

This study did not generate new unique reagents.

Data and code availability

All data of this study are available within the article and [supplemental information](#).

Fabrication of PHI

Metal stamps for PHI are made by CNC Technology. A sheet of foam (Dongguan STC Machinery Equipment) and Teflon paper (Darit Tape) served as the substrate for TPU (matte, Jinda) lamination. In the one-step process, a hot press machine (ST-HS3838, Dongguan STC Machinery Equipment) is set at 120°C to laminate the films for 100 s. After cooling down, an air tube is inserted, and the port is sealed with glue (5500, Ergo).

Device integration and assembly

Dupont thread and electric wire are used to electrically connect all the components, including a microcontroller unit (STM32F103ZE, Zhen Dian Yuan Zi), a battery (NB24V, 82 wh, Yiseneng), MOSFET circuits (GKT16NP, Tikn), air pressure sensors (XGZP6847A, Yujia), miniature solenoid valves (DC24VM, Highen), and a miniature diaphragm pneumatic pump (ZR5551PM, Zhi Rong Hua Guan). Each component is bolted onto the acrylic sheet, and the different acrylic sheets are connected by copper columns. Finally, all equipment is wrapped in an acrylic shell. Pneumatic components are connected through PU tubes (2–6 mm, HEBAI), and air pipes of different sizes are connected through adaptors (6–4,4–2, HEBAI). The interference fit between the 2 mm PU tube and the 0.9 mm Teflon tube (WEICHEN) is achieved by heat shrink tubes (2 mm, WEICHEN).

Simulation

The commercial software COMSOL Multiphysics 6.0 was used to analyze the mechanics of the PHI. In the airflow module, a 2D 10-mm-long pipeline model was established and calculated under the pipeline flow physical field. The material was set to “air,” and the input pressure was set to 2 bar. Parametric sweep is used to calculate outlet flow rates for different pipe diameters. The finite-element method was used in displacement to study the deflection of different-size air capsules under various air pressures. The

simulation results were compared with the experimental results (Figure S9E). In the physics and study tree, the membrane and stationary options were selected to model the system. Only a quarter of the square air capsule is analyzed due to the intrinsic symmetry of the model. The bottom side of the membrane was set as a pressure boundary load to simulate inflation pressure. On the outer circular boundary, a parametrically decreasing edge is applied in the lateral direction as a form of spring support. In the thickness direction, a spring support is added to improve numerical stability. The sequence type is set to physic-controlled mesh, and extra fine mesh is

selected. Under the aforementioned settings of mechanical conditions, deformation and stress can be determined. The elastic modulus (E), Poisson's ratio (ν), and density (ρ) are as follows: $E = 18 \text{ MPa}$, $\nu = 0.4$, and $\rho = 1,235 \text{ kgm}^{-3}$.

Simulation files are available at Zenodo under the DOI <https://doi.org/10.5281/zenodo.10893272>.

Air capsule diameter tests

The output air pressures of the five pressure-limiting valves were calibrated to 20, 40, 60, 80, and 100 kPa, respectively. The input air pressure of the air capsule was adjusted by replacing the pressure-limiting valve. The PHI was attached to a hard board with double-sided tape. When measuring deflection, the focus point of the laser displacement sensor was at the center of the air capsule. When testing force output, the probe (diameter of 2.6 mm) of the force sensor was set to just touch the center of the air capsule. After each deflation, the next round of testing was after the deflection or force to return to the initial value. The test started at 20 kPa, and each air pressure value was measured ten times.

Experiments with human subjects

The experiments with human subjects were performed in compliance with all the ethical regulations under a protocol that was approved by the City University of Hong Kong Research Committee (application no. H002852). Twenty neurologically healthy participants (23–31 years old, median age 26 years, 17 men, 20 right-handed) were recruited from Hong Kong University of Science and Technology, Tencent, and City University of Hong Kong. None of the subjects had read braille before. All of the volunteers gave written informed consent about the experimental procedure. All participants were trained to manipulate the pneumatic-tactile system with the help of experimenters until they understood the sensation of pressure stimulation. To minimize the effect of sensory adaptation, participants were given a 10 min break between trials. The detailed experimental procedures for each experiment are outlined below.

Experiment 1: Minimum tactile perception at different air pressures

This test aims to find the relationship between the perceived tactile sensation and the different combinations of air pressure and air capsule size and serve as the basis for size design. Each size combination has one air capsule.

Volunteers placed their left index finger on the air capsule and did not know the size information. After that, air capsules were randomly inflated under 10, 20, and 30 kPa, and the volunteer was asked if there is a noticeable sensation compared to the deflated air capsule. The process was repeated five times for each volunteer.

Experiment 2: Braille recognition

Before the test began, each volunteer was randomly tested with 5 braille patterns and informed of the correct answers to familiarize themselves with the braille pattern test process. Twenty-six braille patterns for the alphabet were randomly selected and displayed on the PHI (Video S1). The air pressure was 100 kPa. The volunteer was then asked to identify the pattern. Each pattern was displayed twice at most. The process was repeated five times for each volunteer.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.device.2024.100326>.

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AUTHOR CONTRIBUTIONS

Conceptualization, Y.W. and J.L.; investigation, Y.W. and J.Y.; methodology, Y.W., J.L., and J.Y.; experiment, Y.W., J.Y., Y.S., X.H., W.L., and Q.P.; simulation, Y.W., J.L., J.Y., and Y.S.; writing – original draft, Y.W., J.L., and J.Y.; writing – review & editing, Y.W., J.L., W.L., T.Z., Z.Z., X.Y., L.W., and Z.Y.; data curation, X.H.; supervision, Z.Z., Y.G., X.Y., L.W., and Z.Y.; software, L.W.; funding acquisition, L.W. and Z.Y.

DECLARATION OF INTERESTS

The authors declare that no competing interests.

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